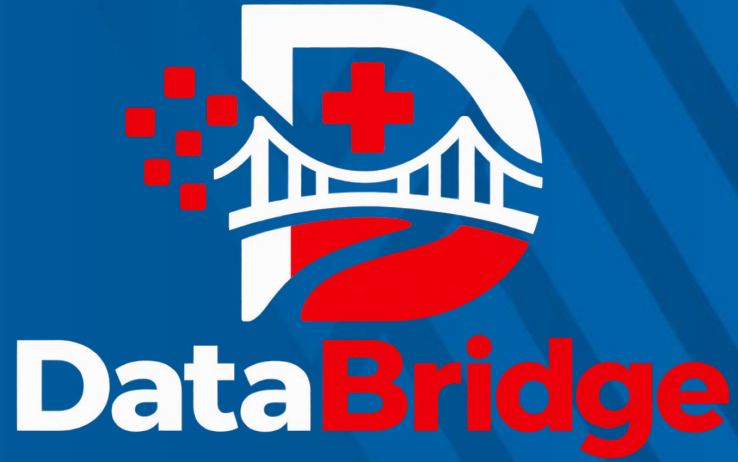
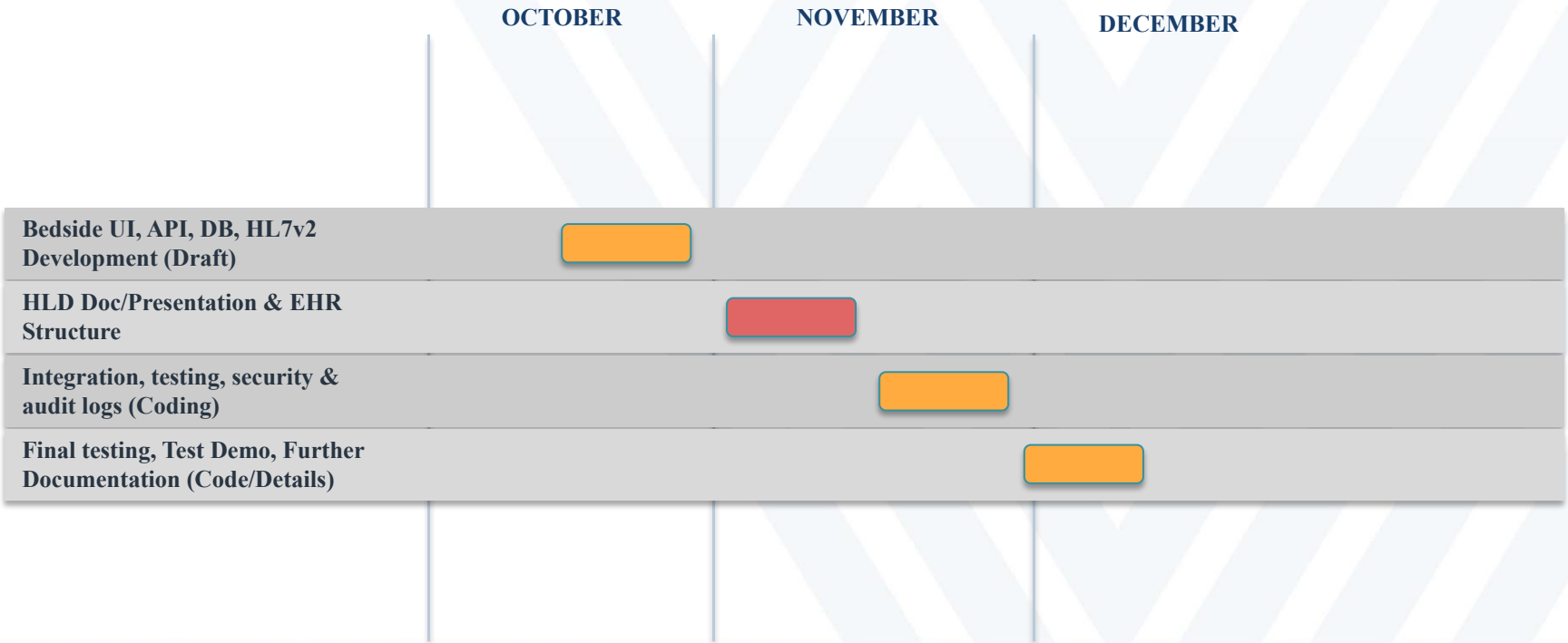
DataBridge High-Level Design (HLD) Phase



By: Nick Fauber, Kayla Granger, Alec Howard, Saph Mabalot

Mentor: Somenath Chakraborty

DataBridge Gantt Chart:



DataBridge

Team 4Health Work Distribution:

Name:	Topic(s):
Kayla Granger	- Dependencies and Assumptions - Security Standards - Gantt Chart
Nick Fauber	- Architecture - Product Functions - Security Considerations
Alec Howard	- Server Operations - Security Considerations - Entity Models / Relations - Technical Data
Saph Mabalot	- Introduction - Document Purpose - Intended Audience and Reading Suggestion

DataBridge

Table of Contents:

1. Introduction

- Document Purpose
- Intended Audience

2. Overview

- System Architecture
- Security Considerations

3. Design Details

- User Perms
- Access Control
- ERD
- Server Ops
- Class Diagram
- Dependencies and Assumptions

4. Technical Data Appendix

- A1→A3

Introduction + Intended Audience

Purpose of the HLD

- Turns SRS requirements into a buildable design
- Defines the data model
- Defines REST API endpoints
- Defines HL7v2 field mapping
- Shows where authentication, RBAC, and audit logging happen

Who This Document Is For

- Instructors and Mentor: full design review
- Development Team: data model, API, HL7v2 mapping
- Testers: permissions and message output
- Nurses and Charge Nurses: their roles in the system
- Admin and IT Staff: security, backup, recovery
- Physicians and Patients: benefit from accurate, private records

System Architecture

Frontend: Browser-based interface for Nurses, Charge Nurses, and Administrators to access patient information and perform authorized tasks.

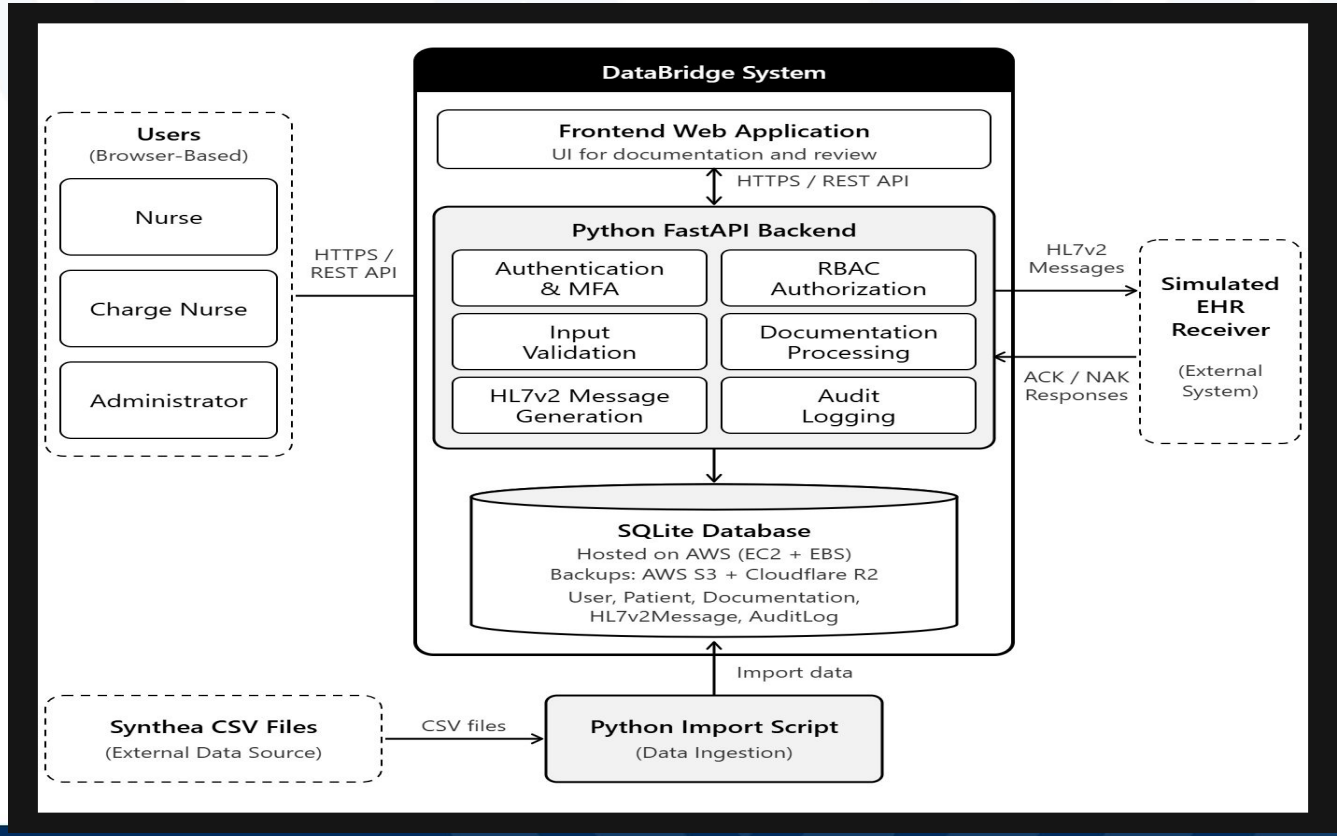
Backend: Python FastAPI handles authentication, MFA, role-based access control, input validation, documentation processing, HL7v2 generation, and audit logging.

Database: SQLite stores user, patient, documentation, HL7v2 message, and audit-log records.

Interoperability: Synthea-generated patient data is imported through a Python script, while validated documentation is converted into HL7v2 messages and sent to an external simulated EHR receiver.

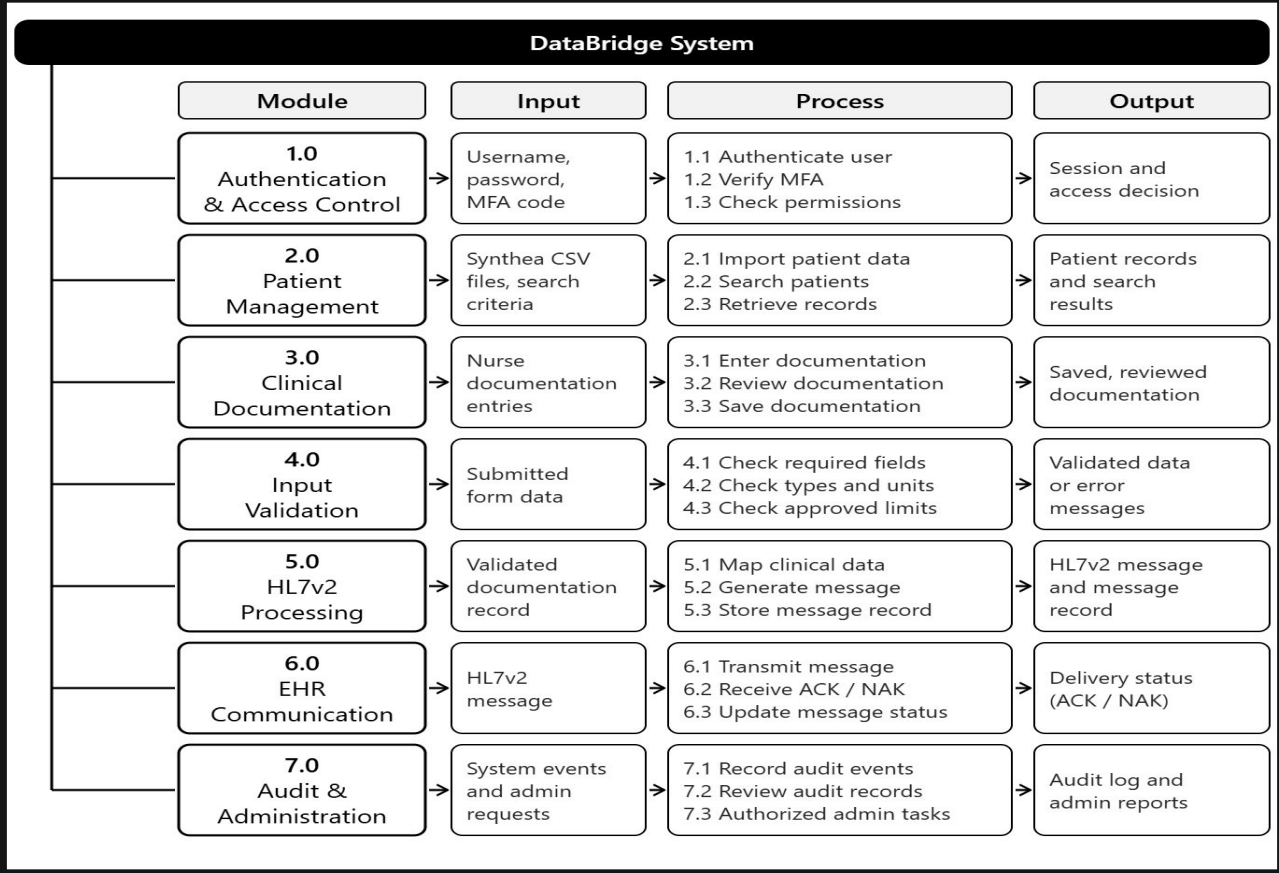
System Architecture

Figure 2.1.1: Architecture Diagram



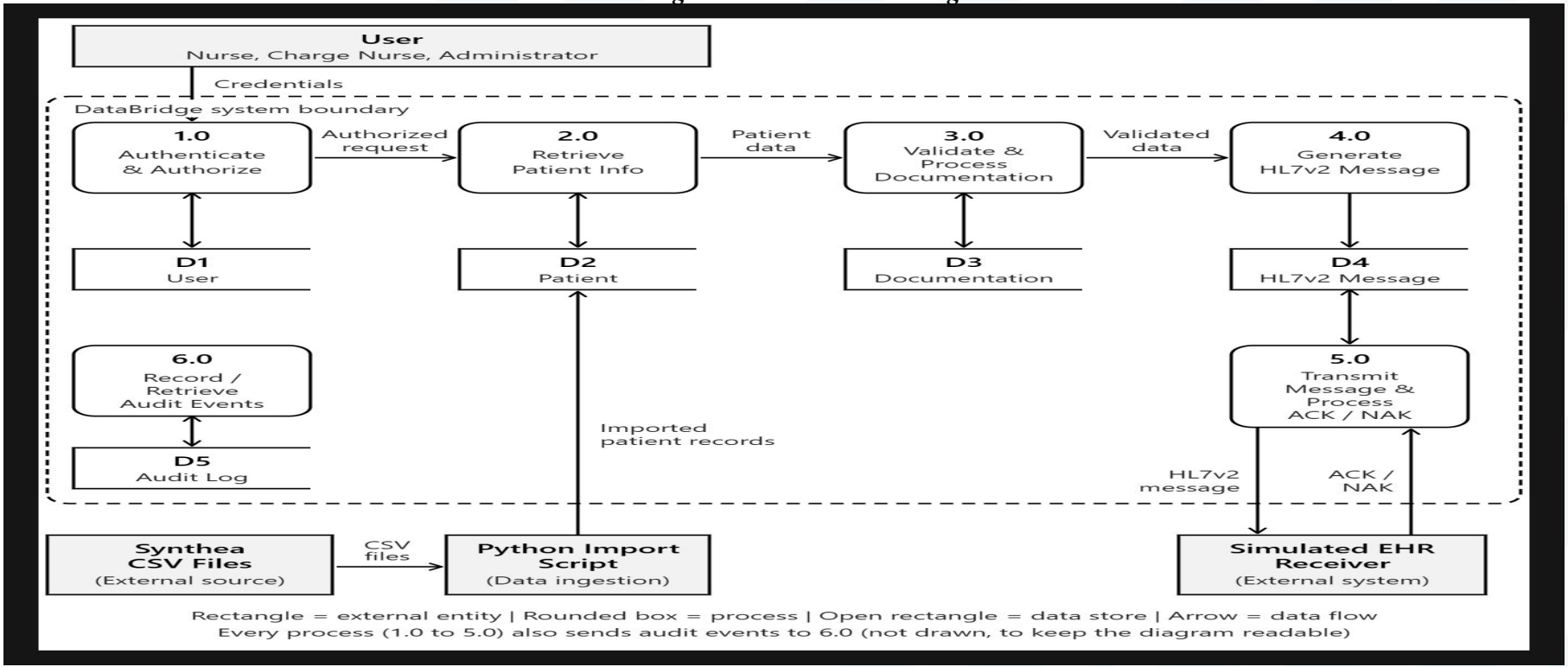
System Architecture

Figure 2.1.2: HIPO Diagram



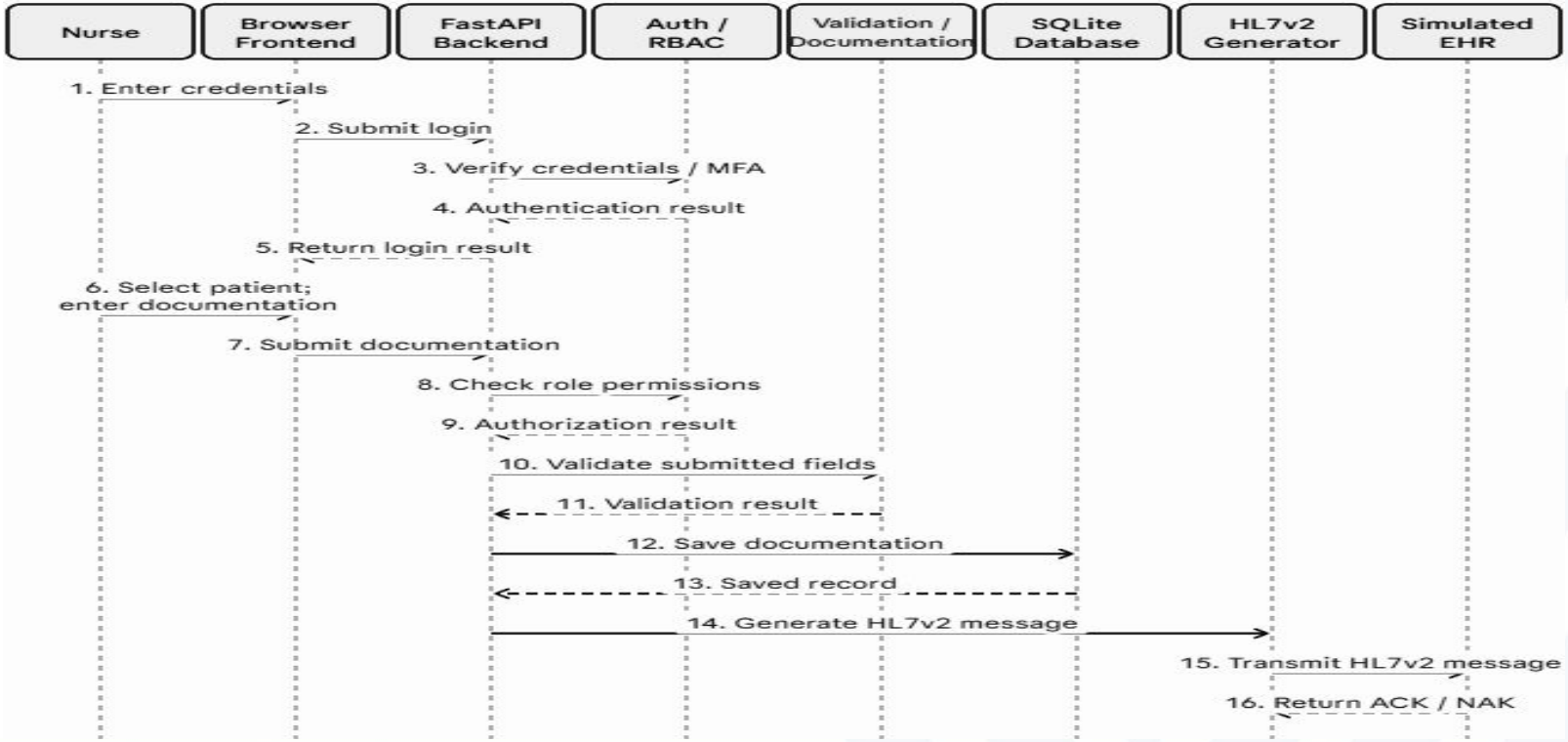
System Architecture

Figure 2.1.3: Data Flow Diagram



System Architecture

Table 2.1.4: High-Level Sequence Diagram



Security Considerations | Engineering Codes & Standards

HIPAA Security Rule	**45 CFR Part 164, Subpart C**	**Establishes security requirements for protecting electronic patient information.**
Administrative Safeguards	**45 CFR §164.308**	**Guides risk management, user responsibilities, and workforce access procedures.**
Physical Safeguards	**45 CFR §164.310**	**Guides protection of tablets, workstations, devices, and physical access to the system.**
Technical Safeguards	**45 CFR §164.312**	**Supports authentication, access control, audit logging, data integrity, and secure transmission.**
Security Documentation	**45 CFR §164.316**	**Establishes requirements for maintaining security policies, procedures, and documentation.**

Applicable HIPAA requirements can be found in 45 CFR Part 164, Subpart C.

Design Details - RBAC

Nurse – Primary User

- Access assigned patients.
- Enter vital signs and bedside observations.
- Submit clinical documentation.
- Receive submission confirmation.
- No access to administrative functions or audit logs.

Charge Nurse – Secondary User

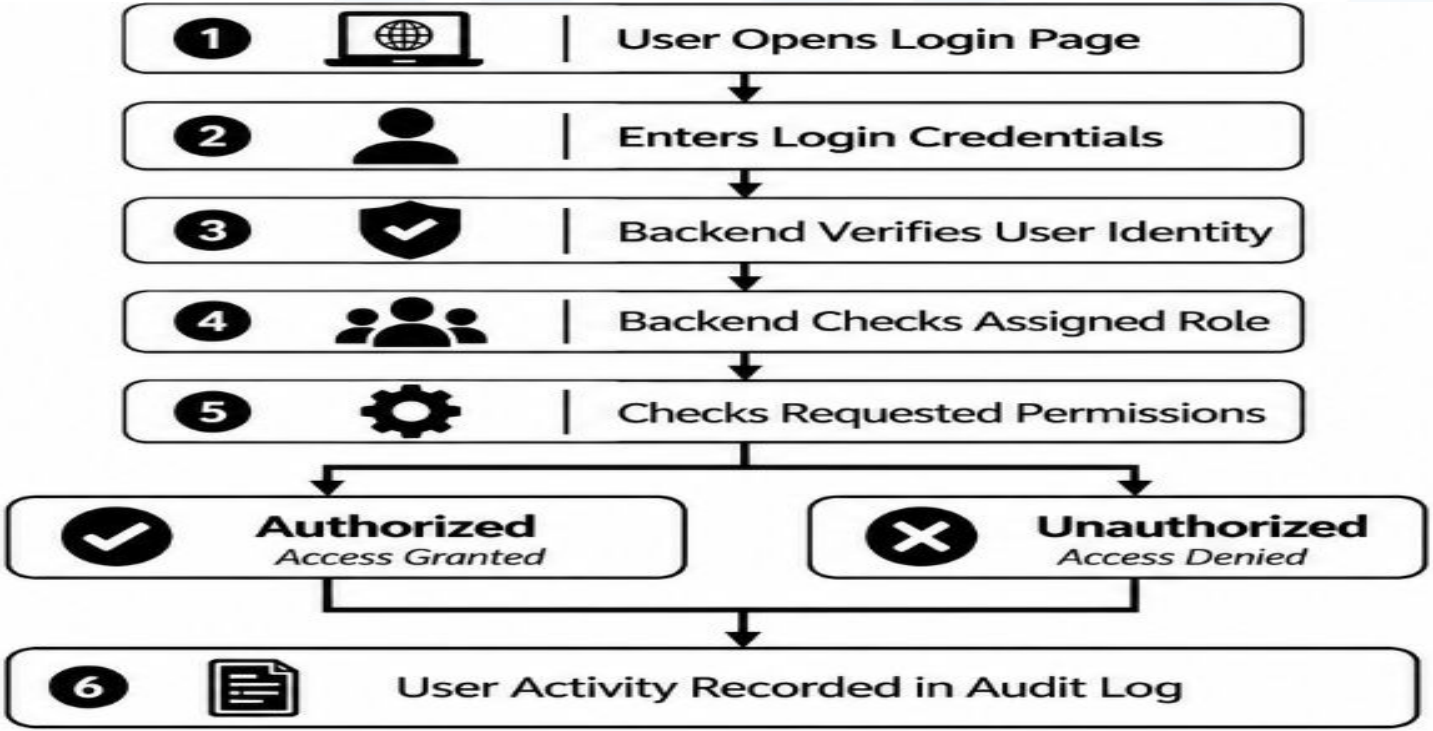
- View the unit-wide patient census.
- Access authorized patient records.
- Perform bedside documentation.
- Monitor patients and coordinate nursing activities.
- No unrestricted administrative privileges.

Administrator

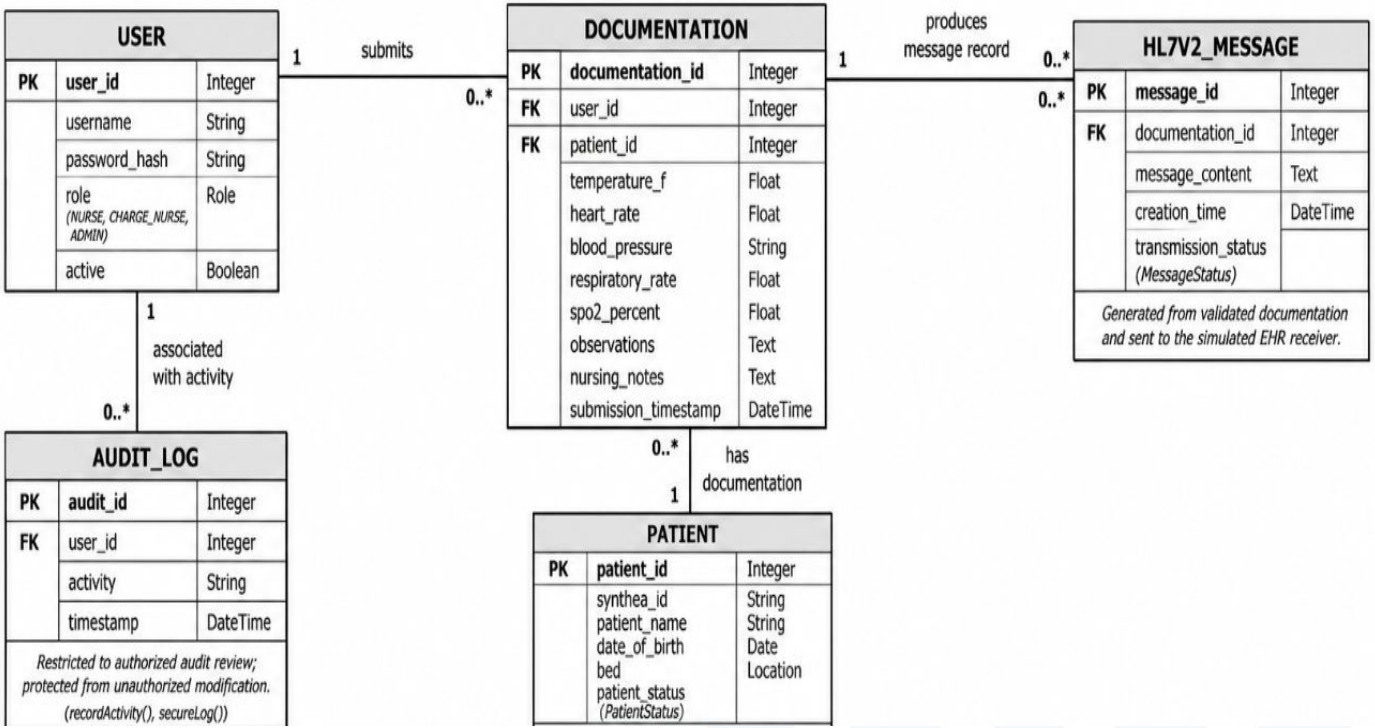
- Assign roles and permissions.
- Manage/Review audit logs and user activity.
- Perform approved system configuration tasks.

System Function / Permission	Nurse	Charge Nurse	Administrator
Authenticate / Log In	✓	✓	✓
View Assigned Patients	✓	✓	–
View Full Unit Census	–	✓	✓
Enter Patient Vital Signs	✓	✓	–
Add Bedside Observation Notes	✓	✓	–
Submit Clinical Documentation	✓	✓	–
Manage User Accounts	–	–	✓
Assign User Roles and Permissions	–	–	✓
Review System Audit Logs	–	–	✓
Administrative System Functions	–	–	✓

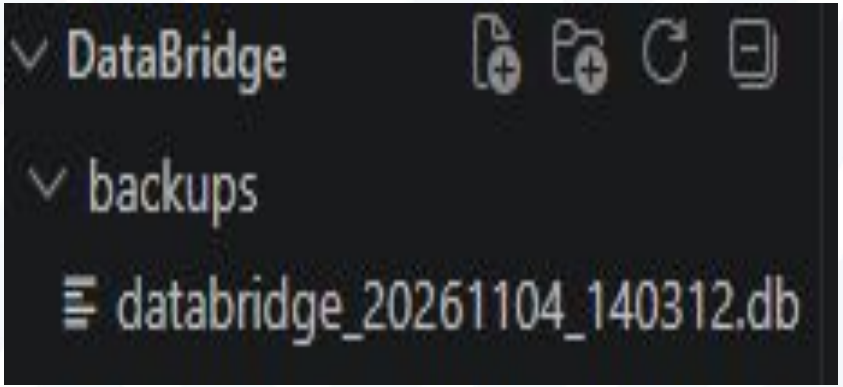
Design Details - Access Control



Entity Relationships - Databridge Architecture



DataBridge Server Operations

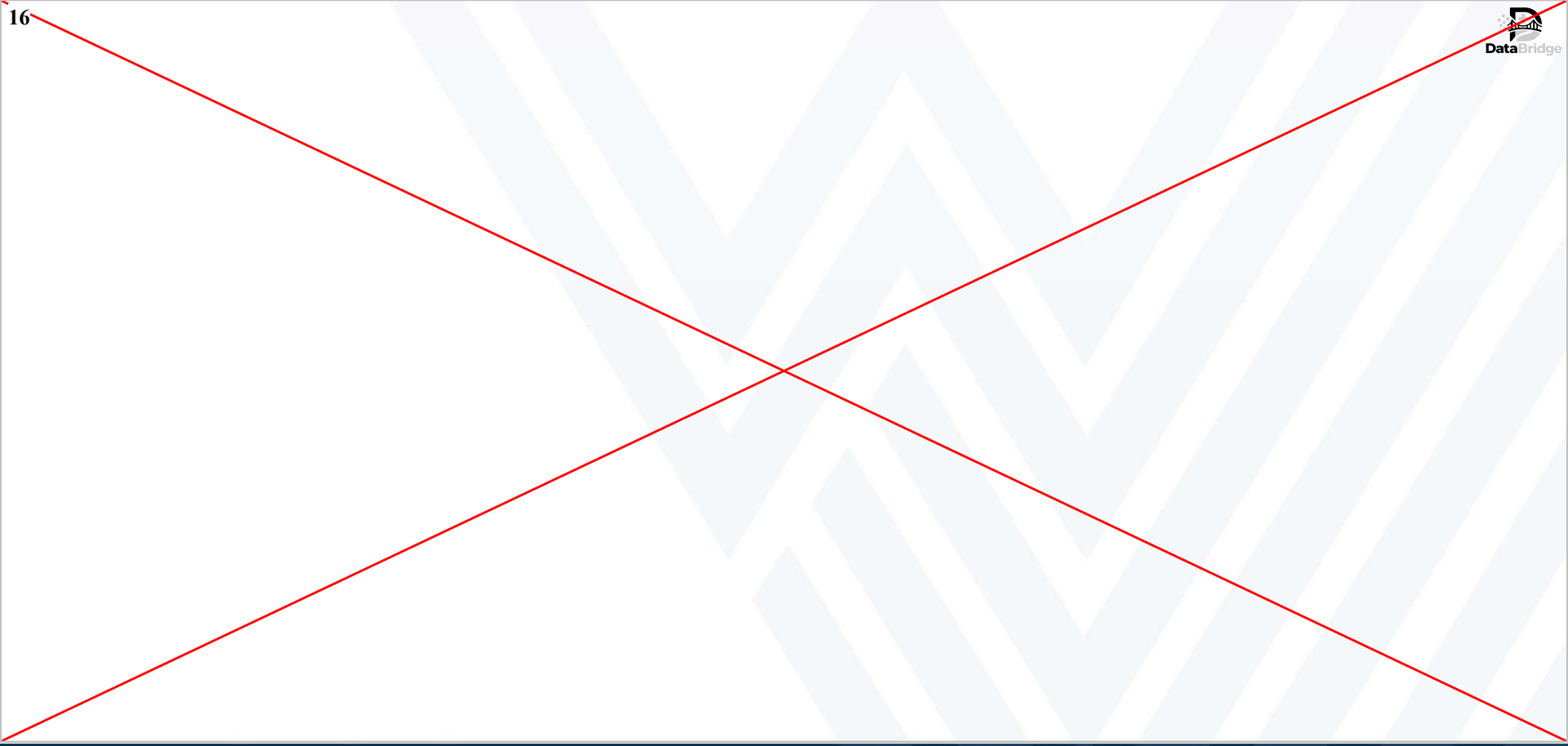


Standard Operations:

- **Role-Based Access Control (RBAC) via web browser.**
- **Submits clinical notes/data to the backend.**
- **Validates data and transmits HL7v2 messages to the EHR.**

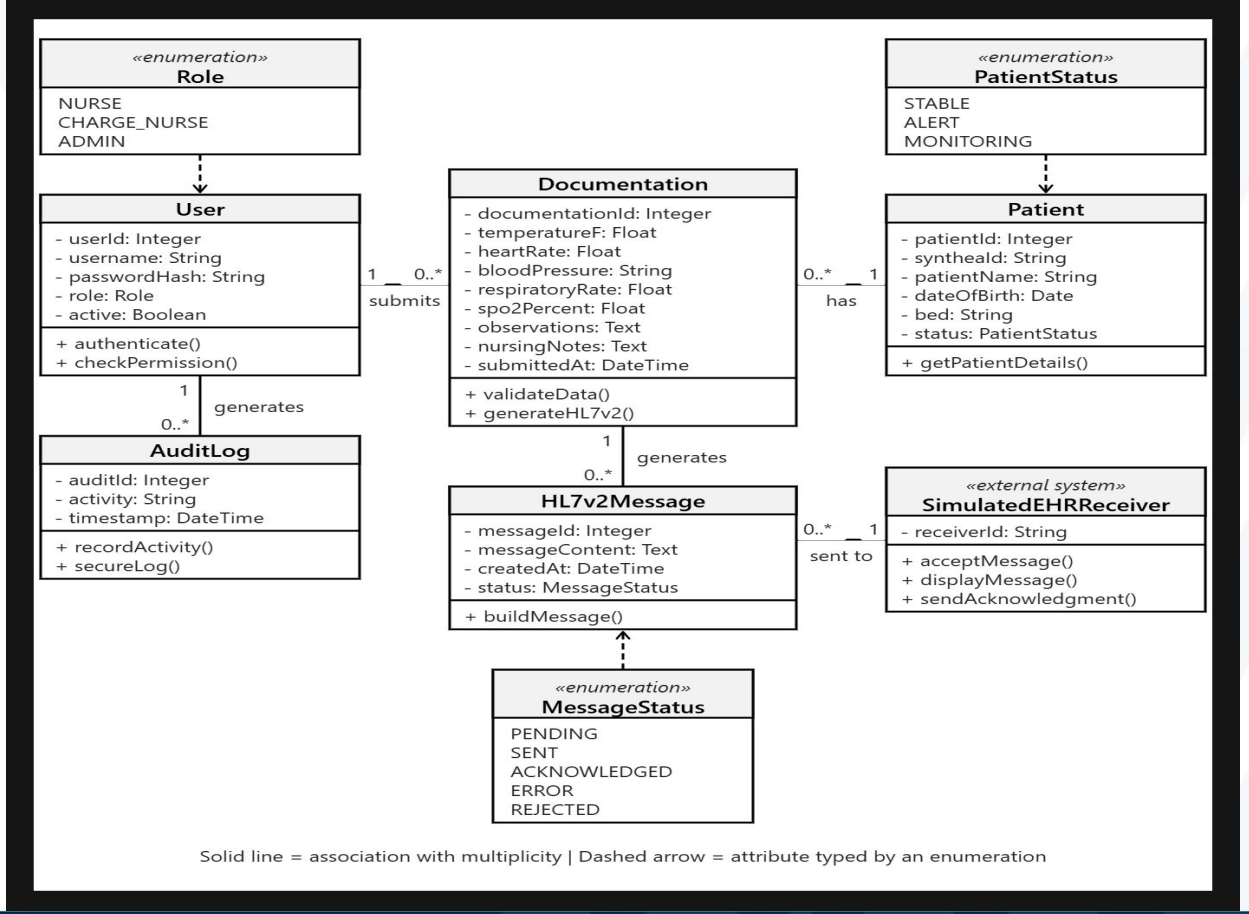
Maintenance & Recovery:

- **Backup: Databases and configuration files are regularly backed up.**
- **Recovery: Check logs, restart components, or restore from backup if needed.**
- **Offline Mode: Users should wait for confirmation before resubmitting notes during an outage.**



Class Diagram (UML)

Figure 3.4.1: Class Diagram



DataBridge

Dependencies and Assumptions

Dependencies

- Backend: Python & FastAPI for validation, authentication, and API communication
- Frontend: React, HTML, CSS
- Simulated EHR: Separate receiver to process & verify HL&v2 messages
- Security: Authentication, RBAC, and encryption
- Patient Data: Synthetic patient records generated through tools like Synthea.

Assumptions

- User Environment
- User Accounts
- Patient Access
- HL7v2 Compatibility
- Synthetic Data
- Project Scope

Technical Data Appendix : A1

Data Model

Entity	Description
User	Stores usernames, password hashes, assigned roles, and account status.
Patient	Stores synthetic patient information imported from Synthea.
Documentation	Stores patient vital signs, observations, nursing notes, and submission timestamps.
HL7v2Message	Stores generated HL7v2 messages and their transmission status.
AuditLog	Records user activities and timestamps for security monitoring.

- One User can submit multiple Documentation records.
- One Patient can have multiple Documentation records.
- One Documentation record can generate multiple HL7v2Message records.
- One User can be associated with multiple AuditLog records.

Technical Data Appendix : A2

REST API Endpoints

Method	Endpoint	Purpose
POST	/api/auth/login	Authenticates users.
POST	/api/auth/logout	Ends an authenticated session.
GET	/api/patients	Retrieves the authorized patient census.
GET	/api/patients/{id}	Retrieves selected patient information.
POST	/api/documentation	Submits clinical documentation.
GET	/api/documentation/{id}	Retrieves a documentation record.
GET	/api/hl7/messages/{id}	Retrieves HL7v2 message status.
GET	/api/admin/audit-logs	Allows administrators to review audit records.

Technical Data Appendix : A3

HL7v2 Field Mapping

OBR stands for Observation/Result.

DataBridge Information	HL7v2 Segment	Field
Message Identifier	MS	MS-10
Message Type	MS	MS-9
Message Timestamp	MS	MS-7
Patient Identifier	PID	PID-3
Patient Name	PID	PID-5
Date of Birth	PID	PID-7
Patient Bed Location	PV1	PV1-3
Vital Signs	OBR	OBR-3 and OBR-5
Measurement Units	OBR	OBR-6
Observation Timestamp	OBR	OBR-14

The simulated EHR receiver returns an acknowledgment:

- ***AA*** **– Accepted: Message successfully accepted.**
- ***AE*** **– Error: Message processing encountered an error.**
- ***AR*** **– Rejected: Message was rejected.**

i.e. Imagine a nurse enters a patient's vital signs into DataBridge.

- **MS: Identifies the message being sent.**
- **PID: Identifies which patient the information belongs to.**
- **PV1: Identifies the patient's visit or hospital location.**
- **OBR: Contains the actual vital signs recorded by the nurse.**

DataBridge | GIT-REPO



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